

Graded, Individual Exercise on Writing Functions and Documenting Tests

Your task is to create your solution for the problem indicated below, creating and implementing your own functions, including creating the test script for each function.

1. Implement the code for function `extractDigits()` [5 pts]
2. Implement the code for function `reverseNumber()` [10 pts]
3. Complete the code for function `display3Digits()` [10 pts]
4. Implement function `sumDigits()` [5 pts]
5. Implement function `isOdd()` [10 pts]

General Instructions and Restrictions:

1. Only function `display3Digits()` and `main()` should have the `printf()` statement as part of the solution.
2. None of the required functions to complete or implement should have `scanf()` statements.
3. You are not allowed to change the function header.
4. Conditional statements (e.g., `if`, `if-else`, `switch`) and other topics or constructs (like ternary operators or arrays) not included in the coverage or in the course notes should not be used as part of the solution.
5. Do not add a `main()` in the `indivExer-LASTNAME.c` file.

Hint/Suggestion:

The function headers are already defined. Requirements per function are indicated. Thus, if you feel stuck on one or feel that the first few functions are more difficult, you are free to work on the other functions first as each will be tested separately (if there is no dependency – one function calling another function).

You are given three (3) files for this problem:

- (1) `indivExer.h` which contains the function prototypes that will be used in the program. This file cannot be modified.
- (2) `indivExer-LASTNAME.c` which contains some initial code that you'll need to complete. Comments indicate the requirement and restrictions imposed for the solution. Refer also to the above instructions and restrictions in this file. This file should not be modified to contain the `main()`.
- (3) `indivMain.c` which contains the `main()` to produce the sample runs indicated.

To debug and test your program using a user-defined module (.h) and with the `main()` in a separate file, you need to do the following:

1. You need to have all three files stated above in the same folder (directory).
2. Then, use the following instruction to compile: `gcc -Wall -std=c99 indivMain.c`
This will produce `a.exe`. Note that there is no need to compile `indivExer-LASTNAME.c` separately.
3. Run in the command prompt as usual. For example, given `a.exe` generated from step 2, just type `a` in the command prompt.

DELIVERABLES: Your C program source file **indivExer-LASTNAME.c** with your own last name as filename. For example, if your lastname is SANTOS, then the source file should be named as **indivExer-SANTOS.c**. Submit also the testScript-LASTNAME.pdf. Upload your source file in AnimoSpace before the indicated submission time.

TESTING & SCORING:

- Your program will be compiled via gcc -Wall -std=c99. Thus, for each function that does not compile successfully, the score for that function is 0.
- Your program will be tested by your instructor with other sample inputs (different from the ones given to you) and with function calls of different parameter values. It is also possible for the instructor to test your indivExer-LASTNAME.c file (the functions) using a different main() program.
- Full credit will be given for the function only if the student's implementation are all correct for all the test values used by the instructor during checking AND only if the student's implementation complied with the requirement and did not violate restrictions. Deductions will be given if not all test cases produce correct results OR if restrictions were not followed.
- In the actual hands-on exam, the test script is not part of the requirement. But, in this exercise, it is included in the scoring [10 pts]. The test script can be created in an excel sheet and saved as pdf. The following are the columns that you need to fill out:

Function Name	#	Test Description	Sample Input (input to the function)	Expected Result	Actual Result	P/F
	1					
	2					
	:::					